Ying-Chun (Thomas) Lee

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EDUCATION

University of Washington, Seattle, WA — ***Sep 2025 – Mar 2027 (Expected)***
Master of Science in Electrical and Computer Engineering
- GPA: 3.98/4.0

Yuan Ze University, Taoyuan, Taiwan — ***Sep 2020 – Jun 2024***
Bachelor of Science in Electrical Engineering
- Overall GPA: 3.73/4.0, Major GPA: 3.94/4.0

SKILLS

- **Relevant Coursework:** ROS, Self-Driving Cars, Software Development, Embedded Systems, Smart Systems, Artificial Intelligence, Deep Learning, Computer Vision, Internet of Things, Cloud Computing
- **Programming**: Python, C, C++, Java, TypeScript, React, HTML, CSS, SQL
- **Tools & Frameworks:** ROS 2, NVIDIA Isaac Sim, Omniverse USD, PyTorch, OpenCV, NumPy, PIL, Docker, AWS, GCP

TECHNICAL EXPERIENCE

Software Development Engineer Intern, *Amazon, Seattle, WA* — ***Jun 2026 – Sep 2026***
- Designed and launched an end-to-end production risk-labeling system spanning Aurora PostgreSQL, Java services on AWS Lambda, 18 REST API operations, and a React/TypeScript UI, replacing recurring schema migrations and deployments with self-service classification management
- Strengthened record-scoped authorization and concurrent-write reliability by deriving access decisions from persisted records and implementing idempotent label writes with explicit transaction handling, constraint translation, and post-transaction reads
- Built keyset-batched bulk labeling for up to 500 risks per request with item-level outcomes and delivered an AWS Cloudscape governance UI with filtering, grouping, drill-downs, and change history; reduced grouped-report API calls from 10 to 1 and data-fetch latency by approximately 80%; and implemented formula-safe CSV/Excel exports

Research Collaborator, *PRIOR Team at Allen Institute for AI (Ai2), Seattle, WA* — ***Oct 2025 – May 2026***
- Collaborated with Ai2 PRIOR researchers to develop real-robot policy infrastructure for deployment, teleoperation, evaluation, and demos across MolmoSpaces, MolmoAct2, MolmoB0T, and TOPReward
- Fine-tuned and deployed pi0, pi0.5, pi0-FAST, ACT, SmolVLA, and Diffusion Policy across Franka FR3, LeRobot SO-100 and SO-101, and YAM robots, with joint position and velocity control and real-time policy inference on the FR3
- Built a multithreaded stack for teleoperation, policy inference, synchronized logging, replay, calibration, validation, and data conversion, reducing the time to collect 100 rollout episodes from about 4 hours to 100 minutes while supporting GELLO and Meta Quest 2 control

Robotics Software Engineer Intern, *CHANG CHUN GROUP – Information Center, Taipei, Taiwan* — ***Jul 2025 – Aug 2025***
Project: ALOHA-ViperX 300S Digital Twin and Imitation Learning Platform
- Created an end-to-end imitation-learning training platform for the ALOHA-ViperX 300S robotic arm, achieving 100% grasp success across 30 simulation trials within a 35–65 cm workspace by integrating a ROS 2/NVIDIA Isaac Sim digital twin with a transformer-based ACT++ control pipeline
- Developed a three-stage trajectory planner with real-time IK constraint validation, workspace-boundary checks, and dual-camera perception for robust simulated grasping
- Automated generation of ACT++-compatible HDF5 datasets from simulation rollouts and documented a reproducible Isaac Sim workflow covering robot control, perception, trajectory generation, and dataset export

PUBLICATIONS

- Shirui Chen, Cole Harrison, **Ying-Chun Lee**, et al., "TOPReward: Token Probabilities as Hidden Zero-Shot Rewards for Robotics," Conference on Robot Learning (CoRL), 2026. arXiv:2602.19313
- Abhay Deshpande, Maya Guru, Rose Hendrix, ..., **Ying-Chun Lee**, ..., Ranjay Krishna, "MolmoB0T: Large-Scale Simulation Enables Zero-Shot Manipulation," CoRL 2026 (Oral). Best Paper Award, IEEE ICRA 2026 Workshop on Synthetic Data for Robot Learning. arXiv:2603.16861
- Haoquan Fang, Jiafei Duan, Donovan Clay, ..., **Ying-Chun Lee**, ..., Ranjay Krishna, "MolmoAct2: Action Reasoning Models for Real-world Deployment," CoRL 2026 (Spotlight). arXiv:2605.02881
- Yejin Kim, Wilbert Pumacay, ..., **Ying-Chun Lee**, ..., Ranjay Krishna, "MolmoSpaces: Large-Scale Open Ecosystem for Robot Manipulation and Navigation," Robotics: Science and Systems (RSS), 2026. DOI: 10.15607/RSS.2026.XXII.091
- **Ying-Chun Lee**, Chih-Yang Lin, Chia-Chun Hsiao, et al., "Using Deep Learning-Based Methods for Automated Segmentation of Soft Tissues from Shoulder Ultrasound Images," IEEE Access, Vol. 12, pp. 111,481-111,492, July 2024. DOI: 10.1109/ACCESS.2024.3432691